I3E Economic Uncertainty Index

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Abstract

We outline the methodology employed in the construction of national and global I3E Economic Uncertainty Indexes. The I3E is a composite indicator designed to quantify the level of economic uncertainty across countries by synthesizing information from four key financial market variables: equity index prices, 10-year government bond prices, exchange rates, and Brent crude oil prices. Each national I3E index aggregates these variables into a single synthetic measure of uncertainty specific to a country's macro-financial environment. The I3E global index is then derived as a GDP-weighted average of the national indexes.

Keywords: Economic uncertainty; Economic index; I3E.

1. Measuring Economic Uncertainty

Economic uncertainty plays a critical role in macroeconomic outcomes, influencing growth, investment, and policy responses. The measurement of economic uncertainty is pivotal for understanding its macroeconomic effects. Approximately one in every 3,000 words in economic reports corresponds to the term "uncertainty" or closely related variants such as "risk" (Ahir et al., 2022).

Uncertainty can be measured. A number of uncertainty indices and empirical approaches have emerged in the last years, each tailored to specific data sources, country coverage, or theoretical underpinnings. These include text-based indices such as the World Uncertainty Index (WUI) (Ahir et al., 2022) and Economic Policy Uncertainty (EPU) (Baker et al., 2016), and statistical measures of variance (Lensink et al., 1999), often derived from autoregressive models or GARCH frameworks.

Volatility indexes like the VIX measuring the expected 30-day volatility implied by stock index option prices like the S&P500 and published by the Chicago Board Options Exchange (CBOE) since 1990 are also frequently used as a measure of financial uncertainty (Whaley, 2009).

We introduce the I3E Economic Uncertainty Index (https://i3e-index.github.io) as an alternative index designed to measure uncertainty. A key advantage of the I3E lies in its ability to be computed efficiently on a daily basis using publicly available financial data available in most developed economies. This reliance on widely accessible and comparable inputs facilitates consistent cross-country analysis of economic uncertainty. The index is designed to have an average of 100 and its values are theoretically bounded between 0 and 200, thereby enabling meaningful comparisons over time and across national economies.

2. I3E Economic Uncertainty Index

The I3E is constructed using the daily closing prices of four financial variables for each of the countries covered:

- Domestic stock index price
- Domestic 10-year government bond price

- Domestic exchange rate
- Brent Spot Price FOB

The specific names of the time-series used for each country are shown in Table 1. Europe Brent Spot Price Free on Board (Dollars Per Barrel) Daily was used for all countries.

Daily growth rates (or returns) of these four economic-financial series $x_i(t)$ are calculated as:

$$y_i(t) = \frac{x_i(t) - x_i(t-1)}{x_i(t-1)}, \quad \text{for } i = 1, 2, 3, 4$$

Next, the volatility of each series is calculated using exponential smoothing (alpha = 0.05) of the squared returns $y_1(t), y_2(t), y_3(t), y_4(t)$:

$$z_i(t) \quad \text{for } i = 1, 2, 3, 4$$

This is the method followed by J.P. Morgan (Riskmetrics) to compute volatility (J.P. Morgan/Reuters, 1996).

Since the resulting series are right-skewed, their natural logarithms are taken. We define:

$$w_i(t) = \log(z_i(t))$$

for i = 1, 2, 3, 4

For the normalization, let μ_i and σ_i be the mean and standard deviation of $w_i(t)$ over the period 1990-2024. The standardized values are:

$$s_i(t) = \frac{w_i(t) - \mu_i}{\sigma_i}$$

The composite standardized series is:

$$S(t) = \sum_{i=1}^{4} \frac{w_i(t) - \mu_i}{\sigma_i}$$

Its standard deviation σ is:

$$\sigma = \sqrt{4 + 2\sum_{i<j} \rho_{ij}}$$

where ρ_{ij} is the correlation between $w_i(t)$ and $w_j(t)$.

The I3E uncertainty index for a country at time t is then defined as:

$$I(t) = 100 + \frac{25}{\sigma} \sum_{i=1}^{4} \frac{w_i(t) - \mu_i}{\sigma_i}$$

During the reference period, the standard deviation (σ) for each country ranged between 2 and 3. Under these parameters, the resulting index is normalized to have an average value of 100 and standard deviation of 25. The index will typically fluctuate between 0 and 200 — although it may exceed these bounds under extreme circumstances.

Using the above formula the I3E for each country can be computed using daily data for the four financial variables (Figure 1). Information about the index daily values and historical data can be downloaded from I3E Index.

3. Subindexes

In addition to the I3E index, the four subindexes $I_1(t), I_2(t), I_3(t), I_4(t)$ corresponding to stock market index, government bonds, exchange rate and Brent prices respectively can be computed for each variable as:

$$I_1(t) = 100 + 25 \times \frac{w_1(t) - \mu_1}{\sigma_1}$$

$$I_2(t) = 100 + 25 \times \frac{w_2(t) - \mu_2}{\sigma_2}$$

$$I_3(t) = 100 + 25 \times \frac{w_3(t) - \mu_3}{\sigma_3}$$

$$I_4(t) = 100 + 25 \times \frac{w_4(t) - \mu_4}{\sigma_4}$$

These four subindices for each country also have a mean of 100 and a standard deviation of 25. Needless to say, the overall I3E Index is not the arithmetic mean of these four partial subindices.

4. Global I3E

The I3E Global is a weighted sum of all available national indexes. Let $I3E_i(t)$ be the value of the index for country i at time t, then the I3E global is computed as:

$$I3E_{global}(t) = \frac{\sum_{i=1}^{n} w_i \times I3E_i(t)}{\sum_{i=1}^{n} w_i}$$

where n is the number of countries and w_i is the nominal GDP in US dollars of country i.

Figure 2 shows the evolution of the I3E Global Weighted from 1990 to 2025 for 22 large economies where the required data is readily available: *Australia, Austria, Belgium, Brazil, Canada, China, Egypt, France, Germany, Greece, Israel, Italy, Japan, Mexico, Netherlands, Poland, South Africa, South Korea, Spain, Turkey, UK, USA*. These countries, together, roughly represent 80% of the global GDP.

Over the past four decades, the I3E has exhibited notable spikes in response to major global events, including Gulf War I, the dot.com crisis, the 9/11 terrorist attacks, the downfall of Lehman Brothers, the Eurozone debt crisis, UK's Brexit, the 2016 U.S. presidential election, U.S - China trade wars, the COVID-19 pandemic and the invasion of Ukraine by Russia.

In addition to the GDP-weighted global I3E, we also publish on a daily basis the global I3E computed as the simple average across all countries.

Figure 3 compares the I3E of the 22 countries and the global I3E (GDP weighted). The global I3E and the I3E for USA, China, Germany and Japan are highlighted. The I3E index for the other 18 countries are shown in different shades of grey in the background. Uncertainty spikes tend to be more synchronized within advanced economies and between economies with strong financial and trade links. Thus, time-series correlation is generally high between countries like USA and France (correlation = 0.94) but much lower between USA and China (correlation = 0.4). These differences may point to persistent differences in how uncertainty generates and spreads internationally. This warrants further investigation in future research.

5. I3E Forward Index

The standard I3E index is constructed using daily realizations of four financial variables: the stock market index, government bond yields, the exchange rate, and Brent crude oil prices. Specifically, it is based on variations in the observed values of these series.

By contrast, the I3E Forward is constructed using market-implied measures of uncertainty for the same four underlying variables. It is computed on a daily basis using the implied volatility derived from one-month constant-maturity options written on each of the four underlying series. As such, the I3E Forward provides a forward-looking measure of economic uncertainty, reflecting market expectations of volatility over the next 30 days.

Implied volatility is not observed directly; it is inferred from option prices. For each underlying series, it takes the market price of a one-month constant-maturity option (or an interpolated constant-maturity price derived from nearby maturities) and compute the value of volatility that makes a standard option pricing model match that observed market price. In other words, implied volatility is the parameter σ that, when inserted into the pricing formula, reproduces the observed option premium. This makes implied volatility a forward-looking measure: it reflects the uncertainty that market participants price over the coming 30 days under the risk-neutral measure.

Concretely, for each underlying variable we invert the Black–Scholes(-Merton) pricing equation. For a European call option with spot price S_t, strike K, risk-free rate r, time to maturity T (in years), and volatility σ, the model price is:

$$C_t = S_t\, N(d_1) - K e^{-rT} N(d_2),$$

where

$$d_1 = \frac{\ln\left(\frac{S_t}{K}\right) + \left(r + \frac{1}{2}\sigma^2\right) T}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T},$$

and $N(\cdot)$ denotes the standard normal cumulative distribution function. The implied volatility $IV(t)$ is the value of σ that solves $C_t^{\text{model}}(\sigma) = C_t^{\text{mkt}}$ (analogously for puts).

Let us call the annualized implied volatility (one-month constant maturity) of each series as $IV_1(t), IV_2(t), IV_3(t), IV_4(t)$. Then their natural logarithms are taken. We define:

$$w_i(t) = \log(IV_i(t))$$

for i = 1, 2, 3, 4

Let μ_i and σ_i be the mean and standard deviation of $w_i(t)$ over the period 1990-2024. The standardized values are:

$$s_i(t) = \frac{w_i(t) - \mu_i}{\sigma_i}$$

The composite standardized series is:

$$S(t) = \sum_{i=1}^{4} \frac{w_i(t) - \mu_i}{\sigma_i}$$

Its standard deviation σ is:

$$\sigma = \sqrt{4 + 2\sum_{i<j} \rho_{ij}}$$

where ρ_{ij} is the correlation between $w_i(t)$ and $w_j(t)$.

The I3E Forward uncertainty index for a country at time t is then defined as:

$$IF(t) = 100 + \frac{25}{\sigma} \sum_{i=1}^{4} \frac{w_i(t) - \mu_i}{\sigma_i}$$

The I3E Forward is conceptually comparable to the VIX index, as both measures are derived from option-implied volatilities with a constant one-month maturity. The most well known VIX index is constructed from the implied volatility of S&P 500 index options and provides a forward-looking measure of expected equity market volatility over the subsequent 30 days.

However, an important distinction lies in the scope of the underlying information set. The VIX reflects uncertainty in a single market—e.g., the U.S. equity market—whereas the I3E Forward aggregates implied volatilities from four distinct financial segments: equity markets, government bond markets, foreign exchange markets, and the oil market. By condensing these four dimensions of market-implied uncertainty into a single composite indicator, the I3E Forward provides a broader measure of forward-looking macro-financial uncertainty.

6. Conclusion

The I3E Index offers a practical, data-driven, and internationally comparable measure of economic uncertainty, leveraging publicly available financial market data. Its simplicity makes it a valuable tool for policymakers, researchers, and investors interested in understanding the dynamics of economic uncertainty across countries.

The I3E Forward provides forward-looking measures of uncertainty across multiple geographies, opening new research avenues for research on the discrepancies between expected and realized economic uncertainty.

Another promising avenue for future research is the study of cross-national variations in uncertainty (e.g., USA-China vs. USA-France), which underscores the asymmetric nature of uncertainty generation and transmission.

Our future research will also address questions related to the relationship between I3E and other uncertainty metrics such as VIX, WUI and EPU, and the relationship between uncertainty, investment, and economic growth in the global economy.

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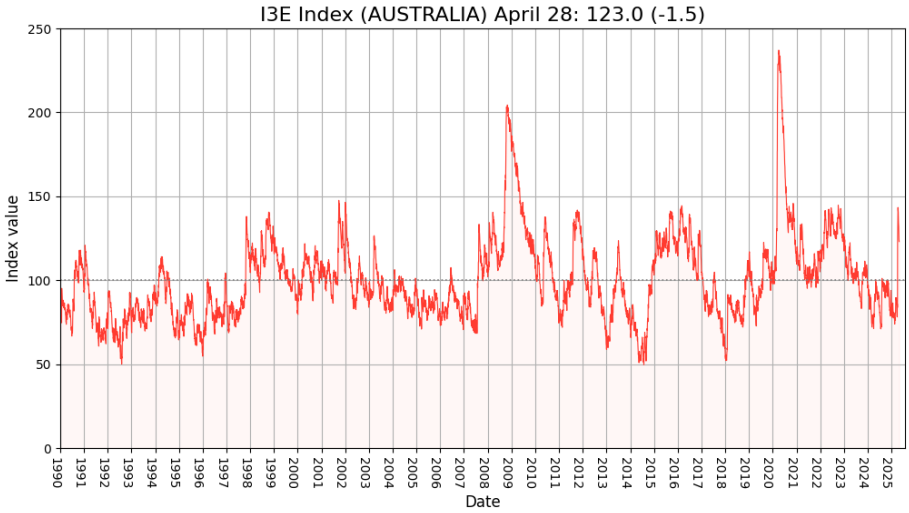

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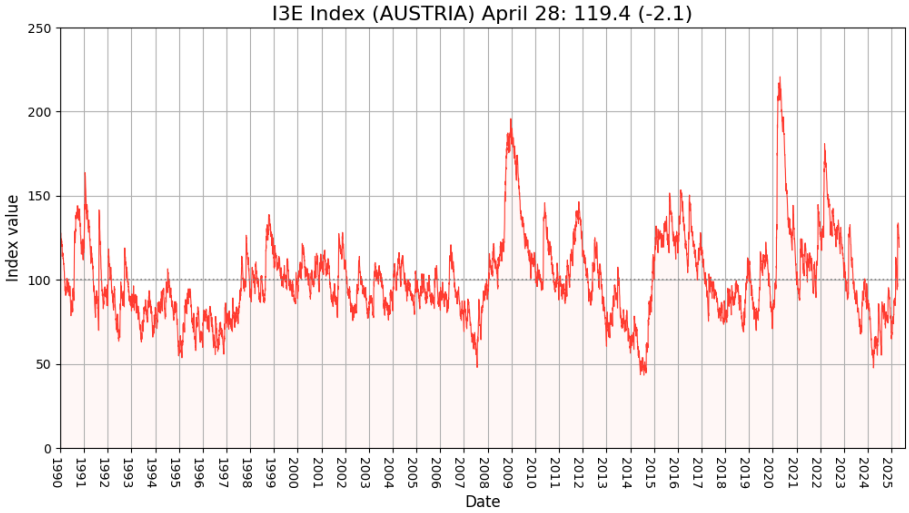

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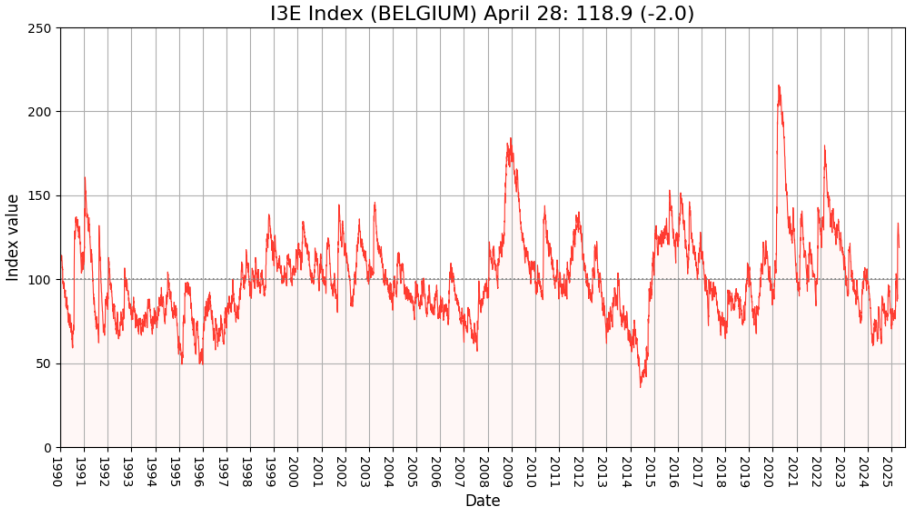
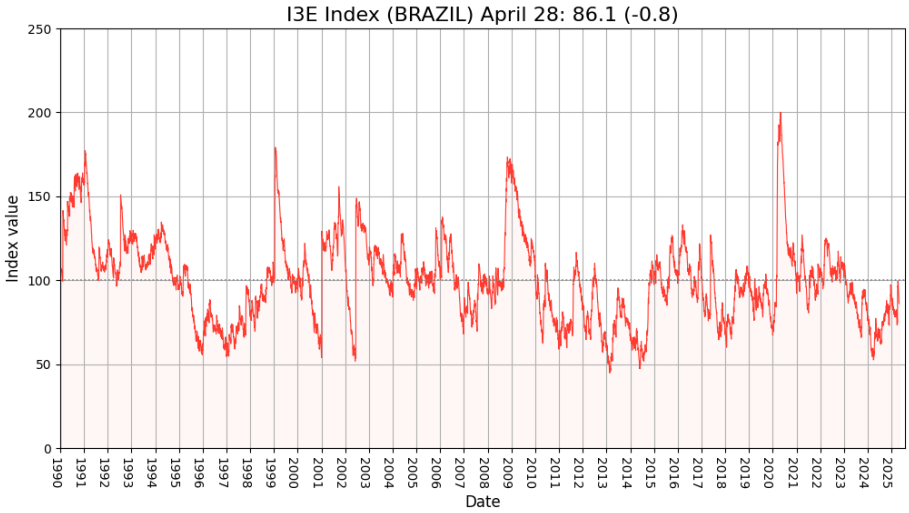
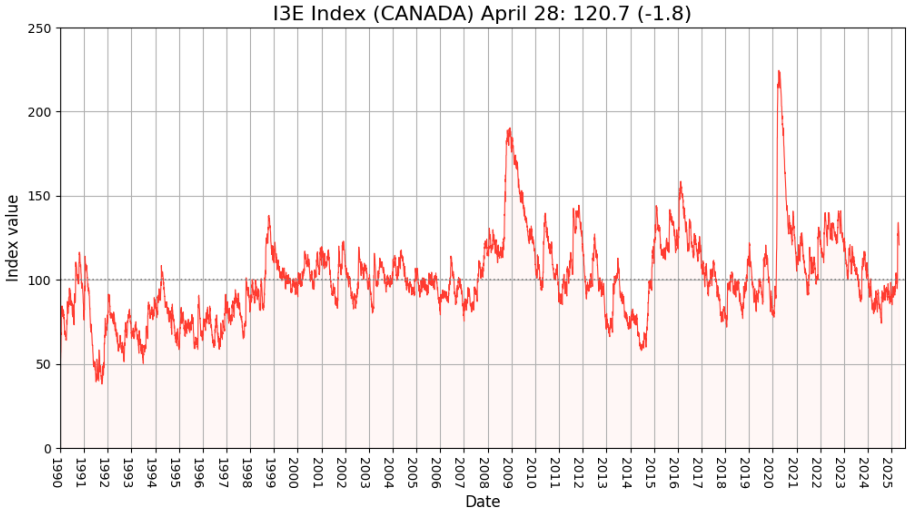


BELGIUM

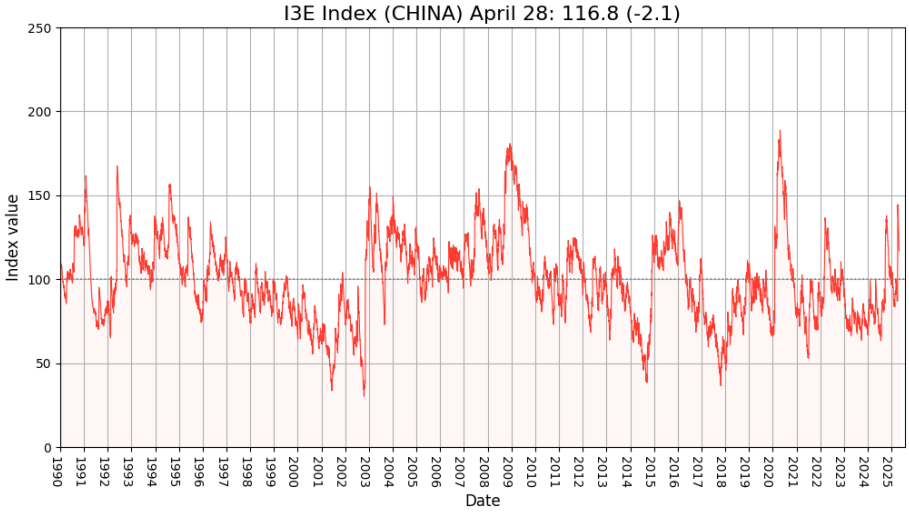
Figure 1: I3E country plots (AUSTRALIA to BELGIUM)



BRAZIL

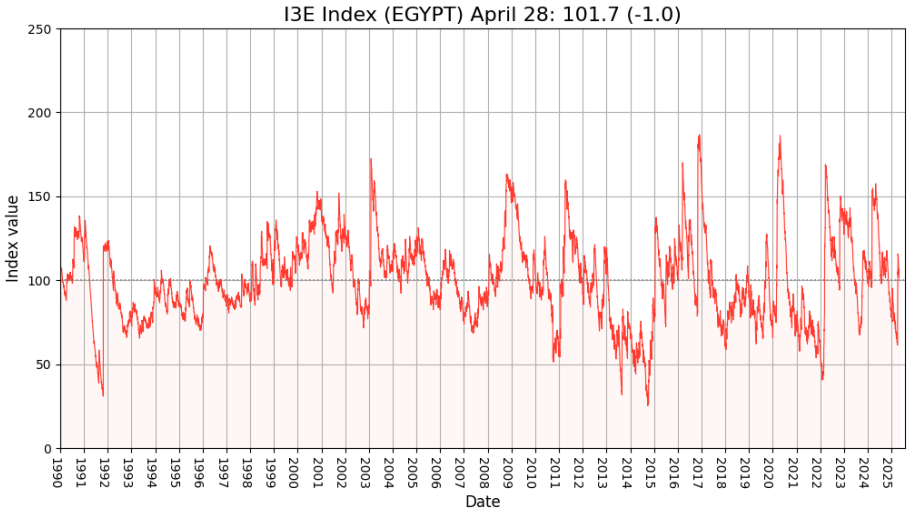


CANADA

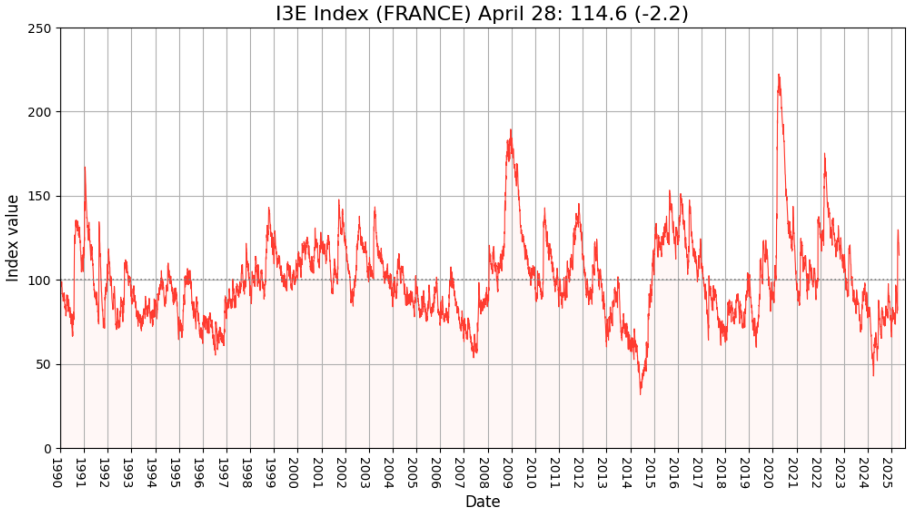


CHINA

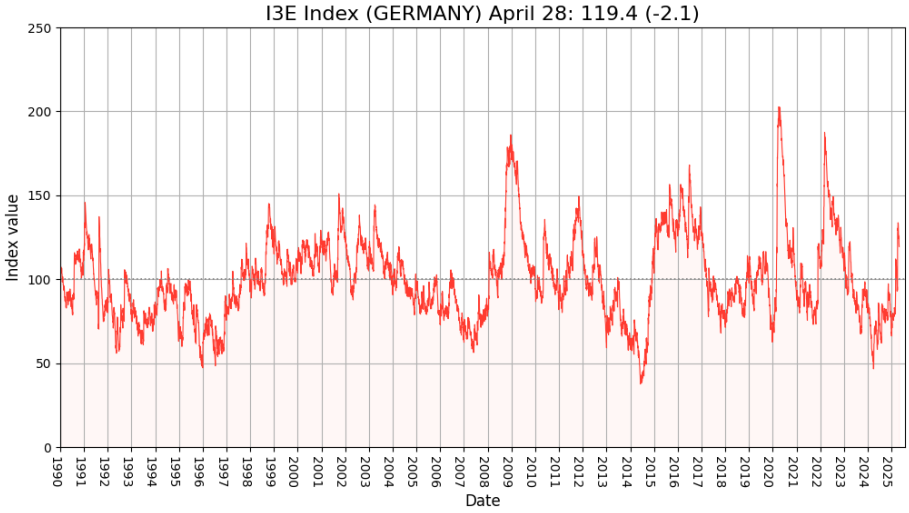
Figure 1: **(continued)** I3E country plots (BRAZIL to CHINA)



EGYPT

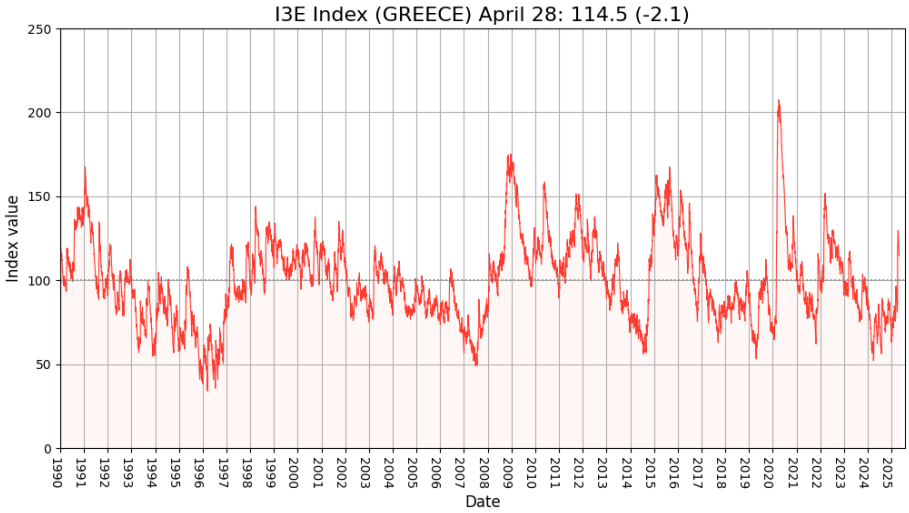


FRANCE

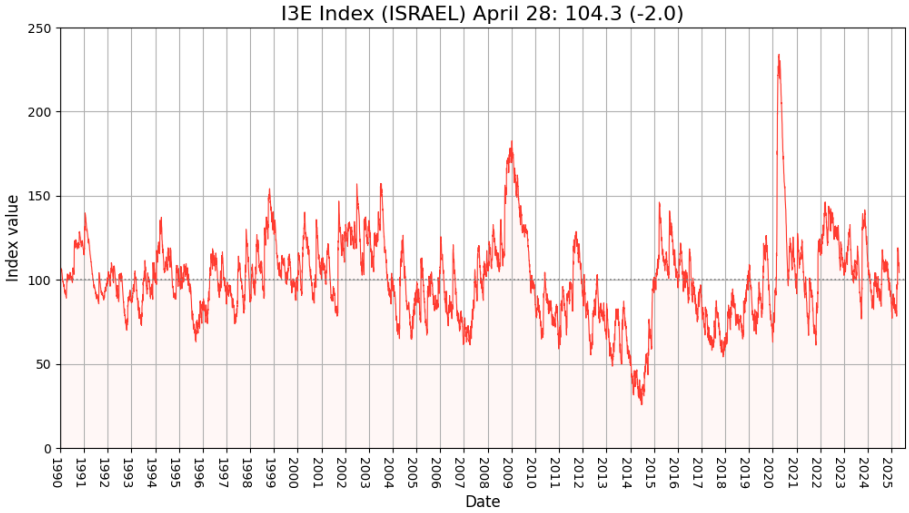


GERMANY

Figure 1: **(continued)** I3E country plots (EGYPT to GERMANY)



GREECE



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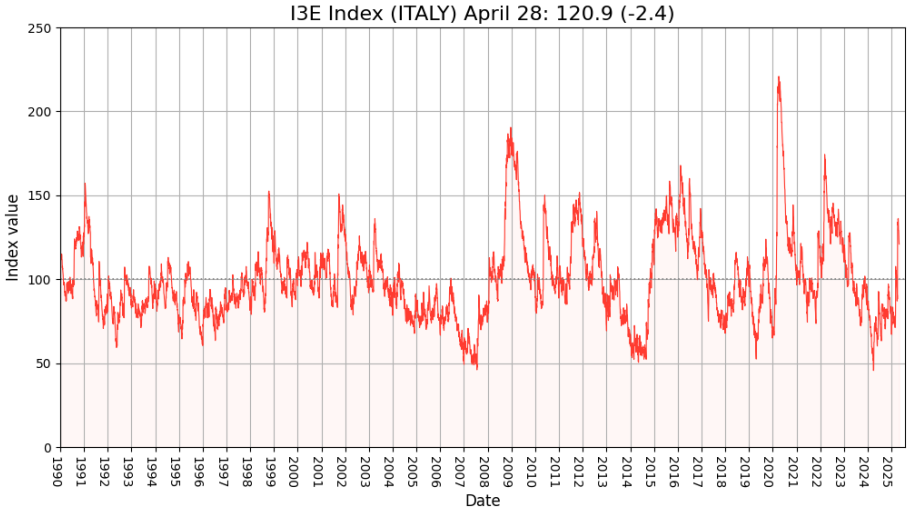
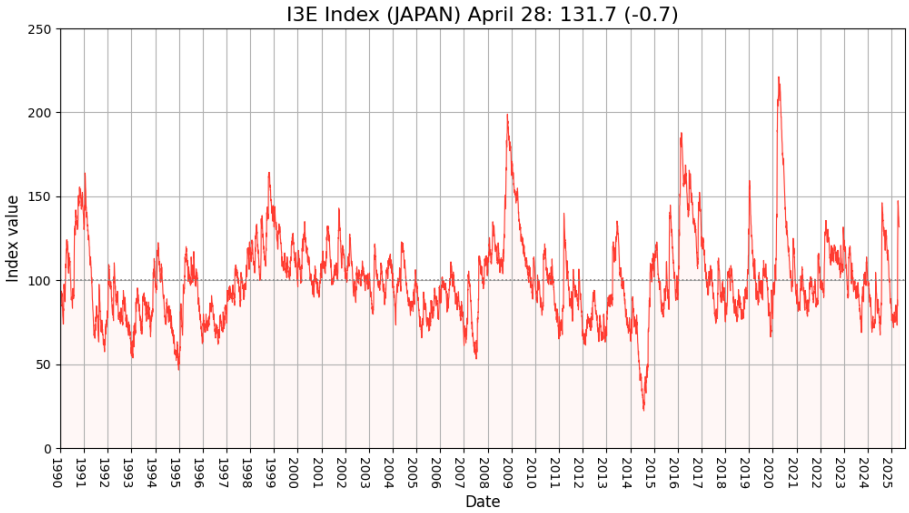
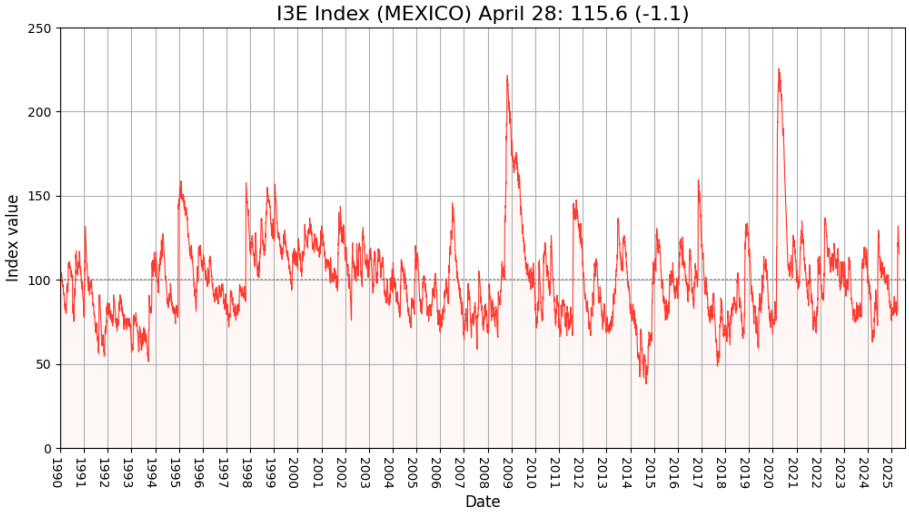


ITALY

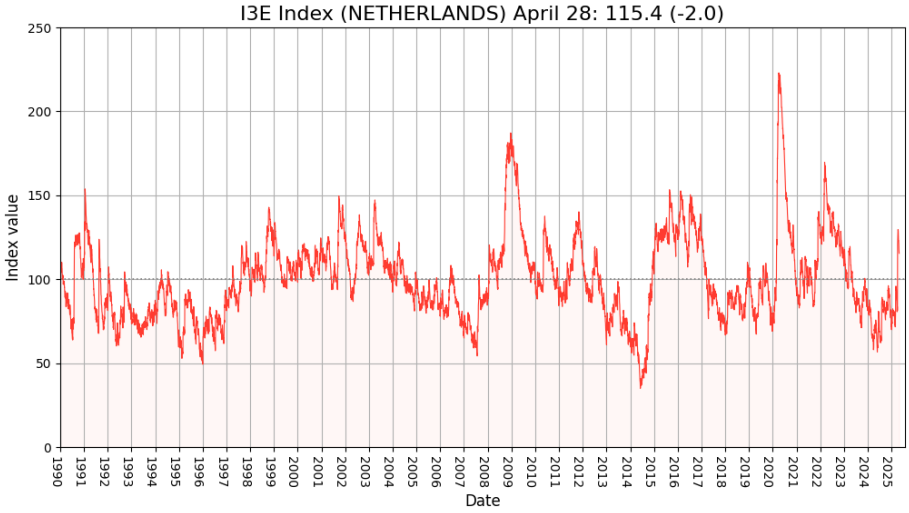
Figure 1: **(continued)** I3E country plots (GREECE to ITALY)



JAPAN

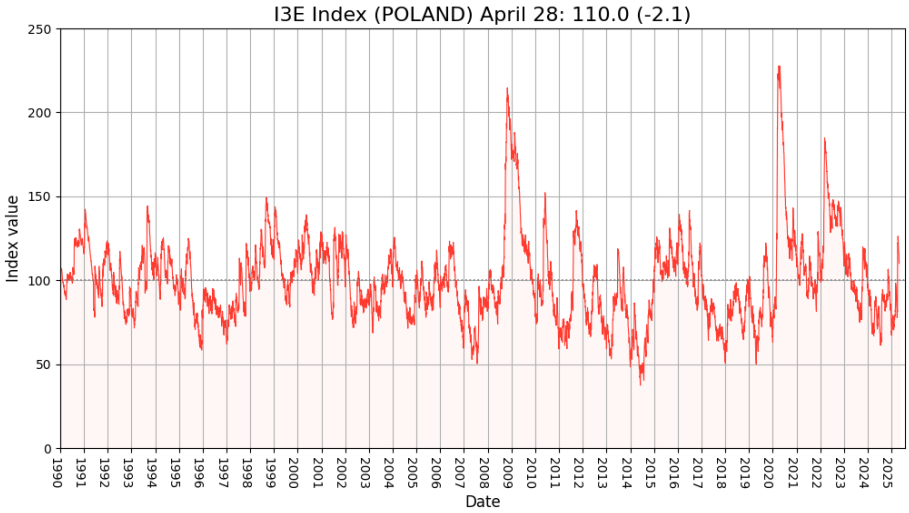


MEXICO

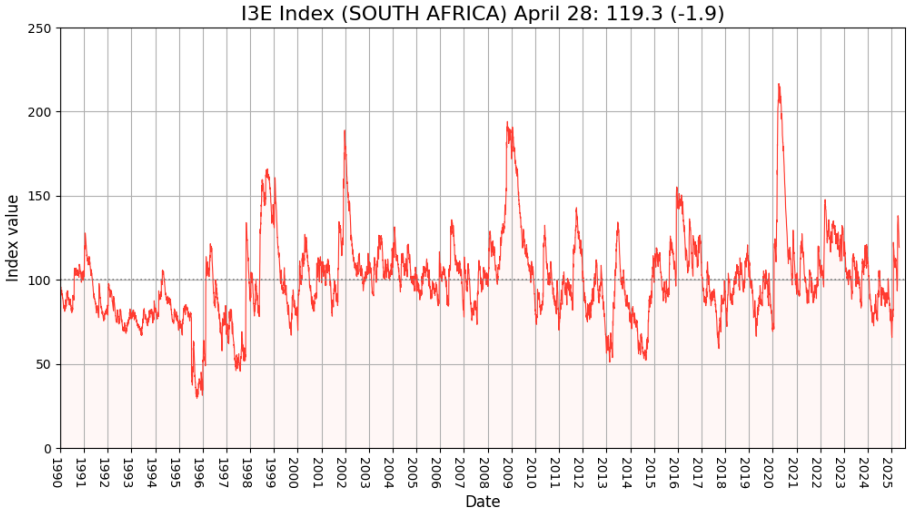


NETHERLANDS

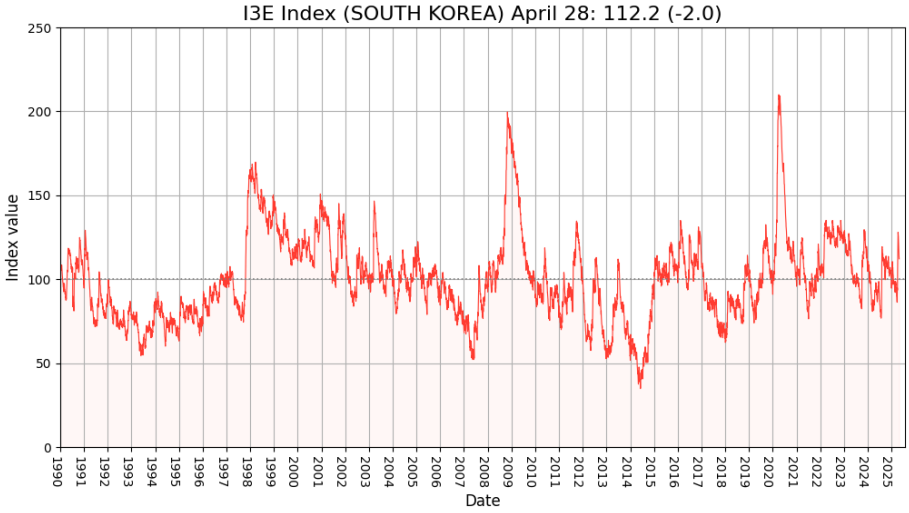
Figure 1: **(continued)** I3E country plots (JAPAN to NETHERLANDS)



POLAND

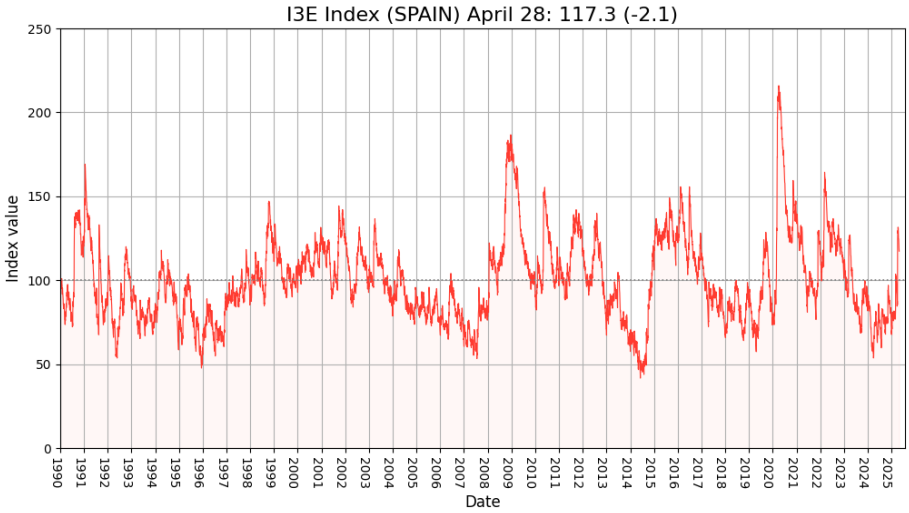


SOUTH AFRICA

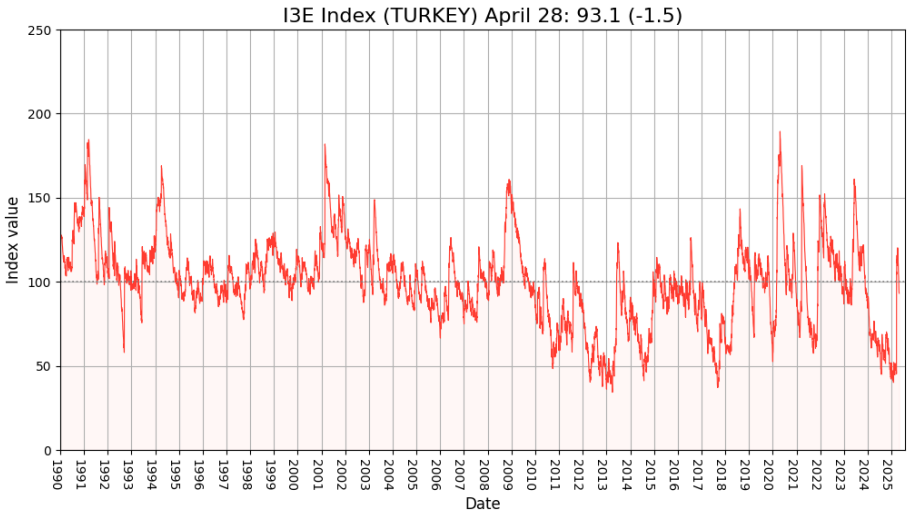


SOUTH KOREA

Figure 1: **(continued)** I3E country plots (POLAND to SOUTH KOREA)



SPAIN



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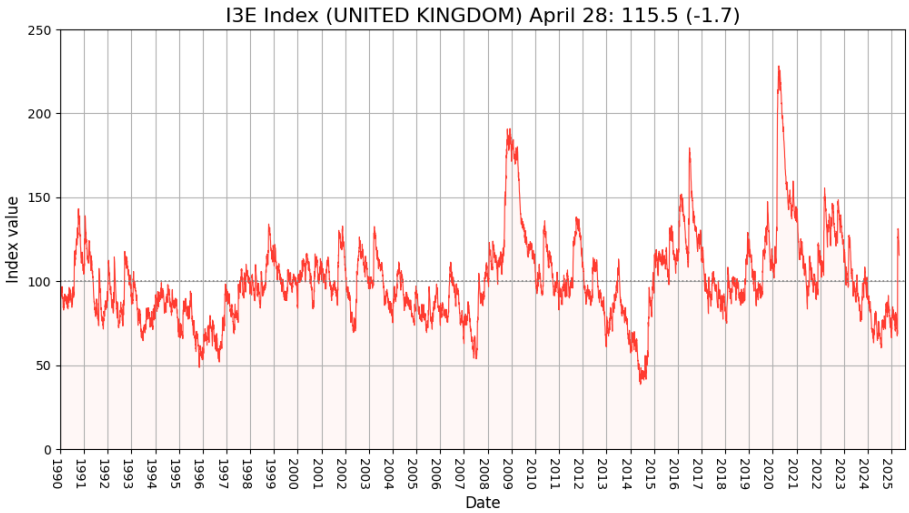
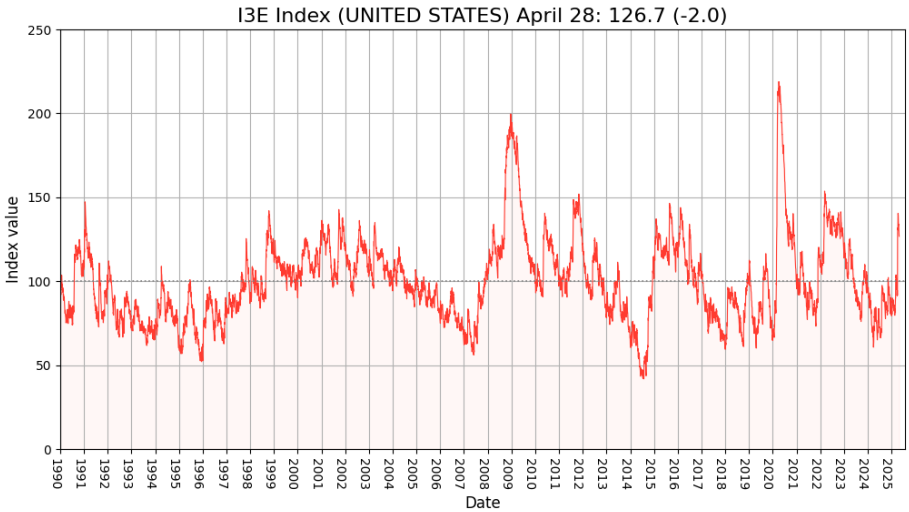


UNITED KINGDOM

Figure 1: **(continued)** I3E country plots (SPAIN to UNITED KINGDOM)



USA

Figure 1: **(continued)** I3E country plot (USA)

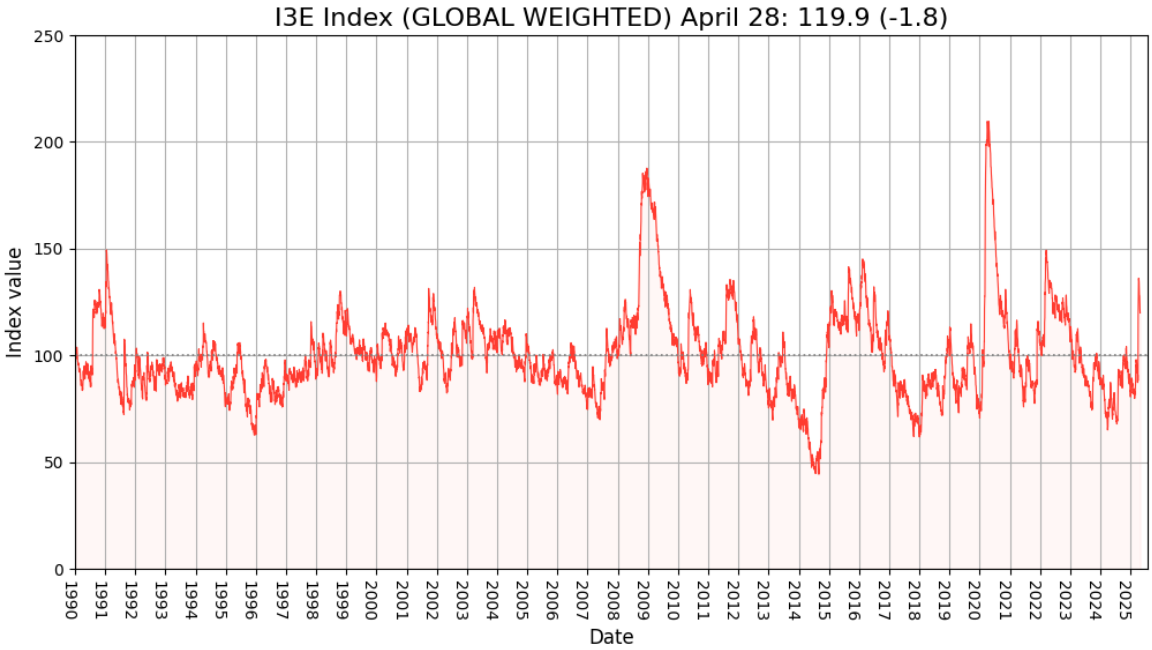


Figure 2: I3E global index

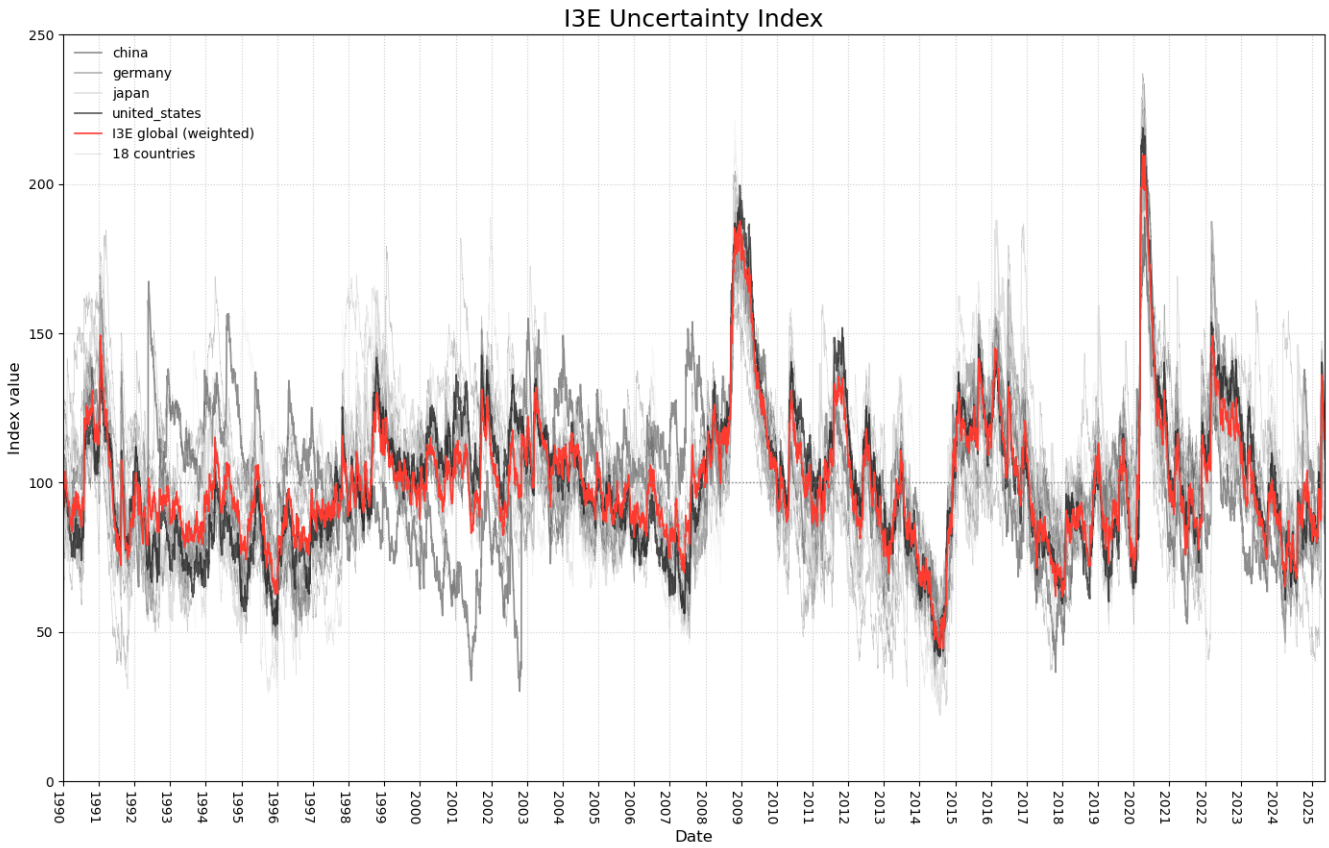


Figure 3: I3E index (all countries)

Table 1: I3E Financial indicators: Datastream Mnemonics

Country	**10Y Bond**	**Stock Index**	**Exchange Rate**	**Since**
Australia	TRAU10T	ASX200I (S&P/ASX 200)	TDAUDSI (Australian Dollar)	1992-05-29
Austria	TROE10T	ATXINDX (ATX)	EUDOLLR (Euro)	1990-01-01
Belgium	TRBG10T	BGBEL20 (BEL20)	EUDOLLR (Euro)	1990-01-02
Brazil	TRBR10T	BRBOVES (Ibovespa)	TDBRLSP (Brazilian Real)	1998-12-11
Canada	TRCN10T	TTOCOMP (S&P/TSX)	TDCADSP (Canadian Dollar)	1990-01-01
China	TRCH10T	CHSCOMP (Shanghai Composite)	TDCNYSP (Chinese Yuan)	2002-06-03
Egypt	TREG10T	EGCSE30 (EGX 30)	TDEGPSP (Egyptian Pound)	2010-08-18
France	TRFR10T	FRCAC40 (CAC 40)	EUDOLLR (Euro)	1990-01-01
Germany	TRBD10T	XETRDAX (DAX)	EUDOLLR (Euro)	1991-07-01
Greece	TRGR10T	GRAGENL (ATHEX Composite)	EUDOLLR (Euro)	1992-09-18
Israel	TRIL10T	ISTMAOF (TA-35)	TDILSSP (Israeli Shekel)	2002-04-09
Italy	TRIT10T	FTSEMIB (FTSE MIB)	EUDOLLR (Euro)	1997-12-31
Japan	TRJP10T	TOKYOSE (Nikkei 225)	TDJPYSP (Japanese Yen)	1990-01-01
Mexico	TRMX10T	MXIPC35 (IPC)	TDMXNSP (Mexican Peso)	2001-07-31
Netherlands	TRNL10T	AMSTEOE (AEX)	EUDOLLR (Euro)	1990-01-01
Poland	TRPO10T	POLWIGI (WIG)	TDPLNSP (Polish Złoty)	1999-11-25
South Africa	TRSA10T	JSEOVER (JSE Top 40)	TDZARSP (South African Rand)	1995-06-30
South Korea	TRKR10T	KORCOMP (KOSPI)	TDKRWSP (South Korean Won)	2000-10-25
Spain	TRES10T	IBEX35I (IBEX35)	EUDOLLR (Euro)	1991-06-03
Turkey	TRTK10T	TRKISTB (BIST 100)	TDTRYSP (Turkish Lira)	2010-01-27
UK	TRUK10T	FTSE100 (FTSE 100)	UKDOLLR (British Pound)	1990-01-01
USA	TRUS10T	S&PCOMP (S&P500)	EUDOLLR (US Dollar)	1990-01-01

Note: "Since" refers to the first date on which all four time series—10-year government bond, stock index, exchange rate, and Brent crude oil prices—became fully available for that country. Before that date, the I3E was computed with the time series available up to that point.